

Astronomer and mathematician Thales of Miletus is the first person in Europe known to have turned away from mythology and used observation to understand the natural world. He effectively invented empirical science and has since become known as the founder of Western philosophy.

MILESTONES

RADICAL THINKER

Establishes a new scientific way to explain phenomena; founds the Milesian School of philosophy.

BASIC SUBSTANCE

Concludes that everything in the Universe derives from a single substance, namely water.

FLOATING EARTH

Proposes that the Earth is a disk surrounded by water, which solidifies to form the land.

APPLIED GEOMETRY

Uses geometrical principles to calculate the height of one of the Egyptian pyramids.

Thales was born in the affluent city-state of Miletus on the coast of modern-day Turkey during the 7th century BCE. Little is known about his life, and what is known derives from ancient Greek scholars writing centuries later, such as Aristotle (see pp.14–17); the historian Herodotus; and the 3rd-century biographer of the Greek philosophers, Diogenes Laërtius. There are no surviving records of Thales' own work or writings. However, according to available source material, Thales was the first of a new generation of thinkers who established a rational, scientific approach to explaining the natural world.

Forging new paths

During Thales' lifetime, superstition and a fear of the gods dominated Western civilization—catastrophic events were ascribed to the wrath of deities who had to be appeased. Yet in an unprecedented break with convention, Thales sought rational explanations rather than religious ones for natural phenomena. He believed that every event,

Thales was one of the Seven Wise Men of antiquity—a group of intellectuals and orators from the 7th and 6th centuries BCE, esteemed for their wisdom.

“Everything is made of water.”

Thales of Miletus, 7th–6th century BCE

